



## Overview

Finance

Admin

Agent

Customer

# System Overview

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One shared PAYGO CRM for finance, admin, agent, and customer portals.

## Purpose

Bumu Paygo supports products that can be paid by instalments: motorbikes, phones, cookers, lockers, solar lamps, and approved assets.

## Main Design

The browser uses secured Vercel API routes. Supabase stores the shared records. Africa's Talking handles SMS and payment flows configured on the backend.

## Architecture UML

Landing site -> Portals -> Vercel API -> Supabase ->

Africa's Talking

Secrets stay server-side only. The frontend never stores payment provider keys, service role keys, or payout credentials.



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# Portal Map

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Each workspace has its own role, screens, and permissions.

| Portal   | Main Work                                                      | Access                               |
|----------|----------------------------------------------------------------|--------------------------------------|
| Finance  | Collections, commissions, reports, reconciliation, payouts.    | Finance/admin role.                  |
| Admin    | Users, products, branches, screening tracking, fraud review.   | Admin role, capped by configuration. |
| Agent    | Customer onboarding, camera documents, deposit prompts, tasks. | Approved agent role.                 |
| Customer | Activation, balance, payments, notifications, payment history. | Approved customer role.              |

## Access Flow

Login

->

Supabase Auth

->

Role check

->

Portal data



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# Customer Onboarding

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The agent registers the customer and captures the required documents.

1. Agent enters customer identity, product, phone, location, deposit, and repayment details.
2. Agent captures passport photo and ID/passport front and back using camera capture.
3. Agent enters next-of-kin details, including location, gender, ID, occupation, and relationship.
4. Next-of-kin receives OTP message and confirms acceptance.
5. Customer receives deposit payment prompt and later activation OTP after approval.

## Onboarding UML

Agent form

->

Documents

->

Kin OTP

->

Deposit prompt

->

Screening

The app cannot force permanent installation on a customer phone. It can provide PWA install prompts, but uninstall control is handled by device policy or MDM, not a normal website.



# Payment Flow

Customer deposits and instalments are initiated through backend payment routes.

## Payment Sequence



- Deposit is requested by the agent after customer registration.
- Customer later pays from the customer portal.
- Callbacks update payment status, balance, progress, and notifications.
- Provider secrets remain in Vercel/Supabase backend settings.

A Till or Paybill must be connected through Africa's Talking or direct Safaricom Daraja configuration before real money can move.



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# Finance Portal

Finance tracks collections, unpaid balances, reports, reconciliation, and commission payout requests.

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## Dashboard

Shows collections, active accounts, overdue amounts, unpaid commissions, and recent trends from Supabase.

## Commissions

Finance sends payout requests to backend. Backend validates commission, agent, amount, and status before payout.

## Reports

Exports finance reports for collections, customers, payments, and reconciliation.

## Alerts

Shows payment events, failed callbacks, overdue follow-ups, and payout results.



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# Agent Portal

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Agents register customers, capture KYC documents, prompt deposits, and track follow-up.

- Agent account login is role checked against Supabase Auth.
- Customer forms require required fields before submission.
- Camera capture stores media through backend storage APIs.
- Deposit prompt starts only after customer registration data is submitted.
- Agent commissions are created from confirmed qualifying payments.

## Agent Workflow

Login

->

Register customer

->

Capture docs

->

Send prompt

->

Track



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# Customer Portal

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Customers activate with OTP, view account status, and make payment requests.

## Activation

After approval, the customer receives an OTP and follows the activation process.

## Payments

The customer portal requests payment prompts through backend APIs and records confirmed callbacks.

## Notifications

Payment reminders, confirmation messages, and account status updates appear in the portal.

## PWA

The portal can be installed through supported browser app-install prompts.



# Admin Portal

Admin manages configuration, screening visibility, products, users, and fraud monitoring.

- Admin registration is protected by setup code.
- Admin accounts are capped by `ADMIN\_MAX\_ACCOUNTS`.
- Repeated failed login attempts lock access temporarily.
- Screening can be automated while admin tracks exceptions and fraud flags.
- Products and branches are managed centrally for all portals.

Admin should be used for monitoring and control, not for exposing provider secrets or direct database keys.





# Security Model

The system is designed so critical business actions happen through backend-controlled APIs.

| Area           | Control                                                                        |
|----------------|--------------------------------------------------------------------------------|
| Authentication | Supabase Auth tokens are verified server-side before data is returned.         |
| Authorization  | Finance, admin, agent, and customer roles are checked per route.               |
| Secrets        | Service role, Africa's Talking, and Daraja credentials must never use `VITE_`. |
| Database       | RLS is enabled and browser roles are revoked from sensitive tables.            |
| Load balancing | API state is stateless and rate limits are shared through Supabase.            |

Do not place payment keys, API keys, service role keys, or payout credentials in frontend files or browser-visible environment variables.



Vercel

Supabase

Africa's Talking

**Checklist**

Support

# Operations Checklist

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Use this checklist before production use or after database updates.

1. Run the latest `supabase.sql` in Supabase SQL editor.
2. Set `SUPABASE\_URL`, `SUPABASE\_ANON\_KEY`, and `SUPABASE\_SERVICE\_ROLE\_KEY` in Vercel.
3. Set Africa's Talking username, API key, sender ID, product name, provider channel, and callback URLs.
4. Set `PAYMENT\_PROVIDER=africastalking` and `COMMISSION\_PAYOUT\_PROVIDER=africastalking` when using Africa's Talking for payments.
5. Set `CRON\_SECRET` and confirm follow-up cron routes are protected.
6. Test login, activation OTP, next-of-kin OTP, deposit prompt, payment callback, commission payout, logout, and data refresh.
7. Monitor Supabase logs, Vercel function logs, payment provider callbacks, and failed notification queues.

The system should be deployed as stateless Vercel routes with Supabase as the central database. This supports scaling and load-balanced execution.